The Pedagogical Application of Alternate Reality Games: Using Game-Based Learning to Revisit History

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ABSTRACT

The advent of the Internet has been instrumental in producing new Game Based Learning (GBL) tools where education and games converge. Alternate Reality Games (ARGs) are one such GBL tool.

Interactive narrative games that use the Internet as a central communications platform, ARGs challenge players to collaboratively collate a fragmented story. When used for educational purposes, Serious ARGs provide a novel form of GBL that encourages critical thinking, develops problem-solving skills and fosters collaborative learning.

However, the pedagogical application of ARGs is still relatively new.

This article presents a background to ARGs and Serious ARGs. It also outlines the lessons drawn from *Plunkett's Pages,* a Serious ARG that focuses on actual historical events. A selection of evaluation criteria, extracted from the reflections of those who played *Plunkett's Pages* are presented. These criteria are intended to enable novice ARG designers or educators to formatively evaluate an emerging ARG design.

Keywords: Alternate Reality Games, ARGs, Serious Games, Game-Based Learning, GBL

1.0 INTRODUCTION

Alternate Reality Games (ARGs) are cross media narrative-based games that use the Internet as a central communications platform. The interactions of participants drive the progression and direction of the story and play experience. Boundaries between reality and fiction are disguised, as game designers ensure that the characters and scenarios react dynamically to player input. Working collaboratively, players collate a fragmented narrative by deciphering codes and clues.

ARGs challenge players to work collaboratively to untangle an underlying narrative that is woven into the fabric of the real world (Bonsignore et al. 2013, Martin et al. 2006). Game creators, known as *puppetmasters* (PMs), typically use the Internet to initially disperse game clues and monitor the progression of their game, while simultaneously interacting with players, through game characters. Meanwhile, players, operating as digital detectives, uncover an underlying narrative; pooling their resources to solve problems, they piece together the narrative, and discuss the game, typically on *Unforum* (2014b) the primary message board for the ARG community. Games are played out on, and bound by the Internet, and yet, elements seep out into the real world, often becoming part of players' everyday lives. Online, collaborative communities, namely *Unfiction* (2014a) and the *Alternate Reality Gaming Network* (ARGN) (2014) are the driving forces behind all ARGs.

The emergence of ARGs at the turn of the 21st century has provided the public with a stimulating cross media storytelling game. The first of these Internet-based ARGs, *The Beast*, emerged in the spring of 2001. Developed by 42 Entertainment alongside Microsoft, *The Beast* was created as a marketing vehicle for the Hollywood movie *A.I. Artificial Intelligence*. Blending aspects from the movie into a twelve-week, murder-mystery, interactive narrative game, the success of the ARG pioneered the formation of a new game genre.

The features of this game genre are included in Table 1:

ARG GAME FEATURES
This Is Not A Game – TINAG ARGs blur the lines of reality and fiction, asking players to suspend their disbelief and deem what they are experiencing as real, and asking PMs to create 'real life' (and not game) experiences. This is known as the *TINAG* mantra.
Rabbit Holes/Trailheads PMs initially entice players to play an ARG through a series of rabbit holes or trailheads (posters, phone calls, messages, emails, websites, adverts, or packages). Rabbit holes should intrigue and capture the players' attention.
Player/Character Interaction ARG players are active participants in the game and interact with characters through multiple media elements. For example print, telephones, email, websites, chat, radio, television, social media etc. Real world encounters also allow players to interact with characters.
Narrative with White Space White space may be defined as gaps in the underlying narrative, which are developed and influenced by the players through game play.
Agency Agency is the level of control and influence that a player feels they have when playing an ARG.
Collaboration and Community ARGs are created for the hive mind, meaning that one person alone cannot complete a game. Collaborative play, and not competitive play, is promoted within the community.

Table 1: ARG Game Features

A Serious ARG typically uses a real-world storyline that offers credible game experiences, and has a primary objective to impart knowledge. ARGs can be used for deep learning, interdisciplinary multimedia inquiry, and offer rich case studies for media and information literacy (Alexander 2006, Barlow 2006, Bonsignore et al. 2013). Their collaborative nature encourages teamwork, develops interpersonal skills and enriches peer-to-peer learning amongst the players (ARGology 2010, Bonsignore et al. 2013, Fujimoto 2010, McGonigal 2007, Whitton 2011b). Game challenges present opportunities for players to learn experientially from problem-based exercises in high-fidelity and meaningful contexts (Bonsignore et al. 2013, Connolly et al. 2011, Moseley 2008, Whitton 2011b). ARGs also appeal equally to male and female players (Lee 2006). Players act as themselves rather than assuming the role of a fictional character (Bonsignore et al. 2013, Lee 2006), which can prove advantageous as it helps blur the lines of reality and fiction. This helps strengthen the aforementioned TINAG. Moreover, Serious ARGs with clear goals help develop and sustain high levels of player application, engagement and motivation (Carson et al. 2009, Fujimoto 2010, Markovic et al. 2007, Moseley 2008).

Furthermore, the barriers to entry for ARGs are low; the creation of an imaginative narrative, affected by players through everyday multimedia technologies, is not only universally available to most, but is also cost effective (Alexander 2006) in comparison to high-end 3D computer and video games.

Over the course of the past decade a number of Serious ARGs have been created to deliver educational value. These include:

- *Aftershock*: where players were provided with practical knowledge on the effects of an earthquake on the San Andreas Fault in Southern California and guidelines on how to react to such an event.
- *Conspiracy for Good*: where players were tasked to bring down an evil global security firm and build a library for children in Zambia in the real world.
- *Evoke*: where the fictional future governor of Tokyo urges action against food shortage.
- *Superstruct*: where players were presented with global health, humanitarian, IT and energy problems to solve.
- *The Tower of Babel*: an EU commissioned project that builds on educational methodology to help teachers use the Internet within language education.
- *Traces of Hope*: highlighted the plight of civilians caught up in war.
- *World Without Oil*: invited people to collectively explore future-changing action regarding the planet's over-dependency on oil.

In a recent interview with the online game developer magazine, *Gamasutra,* acclaimed ARG pioneer, Elan Lee outlines how Serious ARGs provide players with a collective sense of purpose –

You drop two [strangers] in a room, it's hard to get them to connect on any real level. But if you give them a sense of ownership, a sense of purpose, a goal that they're trying to accomplish and by so doing they must rely on each other, suddenly they become friends very quickly. They have these wonderful experiences to share. Their creativity is encouraged and they make progress based on their ability to interact with each other and the world around them. I have only seen good things come of that…I love using these things (ARGs) for education, for teaching people more about their world, and more importantly, teaching them to create the tools necessary to live better lives and make the world around them better… (Gamasutra 2013).

Despite the examples listed above, the use of Serious ARGs, heretofore, has been limited. There remains a gap in the market for a comprehensive study of what makes a Serious ARG a pedagogically sound, teaching tool. The Serious ARG field is relatively young and there exists little literature on their instructional design models, on their use in schools or universities or on how best to implement them. This research part closes this knowledge gap by reporting a selection of findings from an empirical study that aim to assist game designers in the creation of better quality Serious ARGs.

The main objectives of this article are to outline how the creation of a Serious ARG, titled *Plunkett's Pages,* correlated the game play with the desired learning outcomes, motivated players to learn, and enriched player learning.

2.0 PLUNKETT'S PAGES

The integral part that narrative plays in successful ARGs supports the need for a strong storyline to be at the centre of the experience. From a learning perspective, history is fundamentally about stories. This point highlights the suitability of history as a subject upon which many Serious Games can, and have been based (Francis 2006, McCall 2011).

Plunkett's Pages is a Serious ARG, created from November 2012 to March 2013, which provided twenty-five players (approximately 14-15 years old[1]) with the means to learn about the events surrounding the 1916 Easter Rising through the medium of play. The ARG took place on the DkIT campus in a computer laboratory specifically for the players for the duration of the game play. The researcher and his assistants operated as observers for the duration of the Serious ARG. On Day 1, nine players played, on Day 2, twelve players played, and on Day 3, four players played the Serious ARG. The players were initially split into groups of three – while the Day 3 group comprised of four players. However, each of the groups were encouraged to interact and work collaboratively with other groups in order to complete the game.

In the rabbit hole, a YouTube video blog, the players encounter Lucy Reilly, a 14-year-old Irish schoolgirl. Lucy provides some background detail to her life, describing the plight of her grandfather, Tom, who has been acting rather 'strange' of late, resulting in his admittance to hospital for reasons unknown. Whilst hospitalised, Lucy finds a box in the attic of Tom's house. The box contains numerous artefacts, including a dictaphone that details an audio recording of Tom describing the contents of the box and a hidden secret it contains. The box is presented to the players as they investigate the reasons behind Tom's illness and assist Lucy in her quest to uncover the secret.

After interacting with Lucy through Facebook, and listening to Tom's Dictaphone recording, the players make their way to the ARG's primary website, www.plunkettspages.com (2014). The website is an online repository that contains information about the events surrounding the 1916 Easter Rising, an armed insurrection organised by Irish republicans that changed the course of Irish history. Led by the seven members of the Irish Republican Brotherhood (IRB), the aim of the Easter Rising was to end British rule in Ireland, and establish an independent Irish Republic. Joseph Plunkett, an Irish nationalist, poet and journalist, was one of the seven members of the IRB's Military Council. PlunkettsPages.com houses his hidden writings and diary entries. However, the webpages are encrypted and players are challenged with deciphering codes and clues, delving deep into the characters' worlds to uncover vital information that progresses them through the game. The players are not alone in their quest to retrieve the pages of the diary though, as sinister elements are closing in on the life of Tom. As the story unfolds, and players are successful in uncovering page by page of the diary, they race against time to reveal its secret, or risk it being lost forever.

3.0 RESEARCH DESIGN

Both interpretivist and pragmatic philosophies were adopted for the study. Emphasis was placed on the interpretation of *Plunkett's Pages'* players' reflections of their game experiences, and on the pragmatic philosophy of linking theory and practice.

Both deductive and inductive approaches were used. Deductive reasoning was used to postulate theories that could be tested i.e. the hypothesis that a Serious ARG can merge entertainment and educational values to produce a successful learning experience. *Plunkett's*

[1] Minors (those under the age of eighteen) were the research population for the study. The DkIT Ethics Research Committee approved the research, while the parents or guardians provided their consent.

Pages and its in-game assessments and questionnaires test this hypothesis. An inductive approach was employed when analysing data collected during *Plunkett's Pages*. The theoretical inferences drawn from game data were reflective of the players' views on the factors that lead to successful Serious ARGs. The research methodology primarily used in this study was design-based research. The term describes a particular stance taken to design- and intervention-based studies of learning, with the goal of developing models and understanding of learning in the naturalistic, intentional learning environments (Brown 1992, Hevner 2004, Iivari and Venable 2009, Sandoval and Bell 2004, Wang and Hannafin 2005). Design-based research builds new and innovative artefacts with the aim of achieving improvements and solutions to a particular problem (Iivari and Venable 2009). Today, scholars engage in design-based research to better understand how to orchestrate innovative learning experiences for students in everyday educational contexts as well as simultaneously developing new theoretical insights about the nature of learning. Intertwining research and practice fits well with the purposes of education given its fundamentally interventionist nature. This aspect aligns itself with the pragmatic approach that helped guide this study. The incorporation of an ARG into the realm of the natural learning setting, with an aim of effectively providing the information on the events surrounding the 1916 Easter Rising, links both theory and practice. Offermann *et al's.* (2009) design-based methodology specifically steered the development of *Plunkett's Pages* and the overall study.

In *Plunkett's Pages,* a mixed-method research approach was used to gather data during game sessions. The collection of data from several sources triangulates the findings in ways that they may be mutually corroborated. Data was collected through the following methods:

1. Participant-observation: the researcher adopted the role of the 'observer as participant' (Saunders et al. 2009), observing the game play "without taking part in the activities in the same way as the 'real' candidates" (Saunders et al. 2009, p. 294).
2. Questionnaires: player knowledge of the subject matter was examined prior to, during and after the game. This gathered quantitative data i.e. the players' test results. Knowledge acquisition can be evaluated in Serious Games with knowledge testing (Mayer et al. 2014). A benchmarking system, devised between the researcher and the teachers, maintained consistency and reliability when dealing with the players' pre- and post- game quantitative data in Questionnaire A. Questionnaire C focused on the factors that led to player enjoyment of the game play or conversely didn't, and the factors that contributed to their learning. Finally, Questionnaire E was conducted after the researcher extracted a number of evaluation criteria from the study. This questionnaire was presented to the players in a later time period as part of the validation of the results.
3. Focus groups: were useful in understanding what features of the ARG made the game an enjoyable learning experience for the players. Conducted in the aftermath of the game, all focus groups took place in a relaxed environment, with participants encouraged to share their different perceptions and points of view without the need to reach consensus (Kreuger 1988). Eight focus groups were conducted, involving between 3-4 participants at each session, ranging from 40 minutes – 75 minutes in length.

Thematic analysis, a method for identifying and reporting themes within data in rich detail, was used to analyse the qualitative data. This process helped generate rich and detailed data that can contribute to an understanding of the contextual factors that furnish any learning improvements. This data stems from the players' opinions of their game play experiences, and was primarily concerned with their assertions on the relative factors that enriched their learning. Braun and Clarke's (2006) six phases of thematic analysis guided this process.

4.0 FINDINGS

The results of the comparative analysis indicated that all twenty-five players improved their knowledge. Indeed, all players showed a noteworthy increase in their post-ARG percentage scores with improvements of knowledge in Players 7, 30, 18 and 21 increasing significantly by 600%, 814%, 1020% and 1800% respectively. The improvements in knowledge provide a good indication of the learning benefits of the game.

Thematic analysis extracted evaluation factors from player's considerations of what affected the quality of their experiences. A selection of these evaluation factors, that is, lessons drawn from player's responses, are now presented. They are structured as follows: (1) a short introduction to the theme and sub-themes, (2) player quotes to illustrate the theme and the researcher's interpretation of same (3) an interim evaluation criterion, (4) player responses to the interim evaluation criterion, and (5) a conclusion and presentation of an evaluation criterion extracted from the analysis. The legend for the sources is as follows:[2]

Questionnaire Source: P28QC8
P28=Player 28. QC8=Question 8 in Questionnaire C.
Focus Group Source: P32L178
P32=Player 32. L178=Line 178 of the data set.

The overarching theme of *Focused Game-Based Learning* emerged from a review of the data corpus gathered from *Plunkett's Pages.* In the next sections, insights are provided in relation to how the theme emerged and what the players said[3] that led to its formation. This theme contains four sub themes, *Assessment, Feedback, Collaborative Learning* and *Debriefing*, and their subsequent evaluation criteria.

4.1 Sub-Theme: Assessment

4.1.1 Introduction to Assessment

This first sub theme deals with the aspect of assessing player performances in Serious ARGs. In *Plunkett's Pages,* player learning primarily centred on their ability to overcome the challenges presented by in-game assessment.

4.1.2 Findings and Interpretation of Assessment

Challenges and assessments provoked and stimulated thought amongst players, prompting them to question aspects of the game and events from Easter 1916, and rewarding their performances.

[2] Note that sources prefixed with the letter 'E' (i.e. EP23L102) relate to player reflections from follow-up questionnaires (Questionnaire E) and focus groups where the presentation of the interim findings were discussed.

[3] The language has been tidied slightly to improve readability, but is informal as it remains very close to the source.

…the grids and the crosswords gets your mind thinking – you have to think about different things – its challenging – it gets you thinking. If everything was presented to us there and then, we probably wouldn't have enjoyed it as much because it's rewarding as you figure things out and get through to the next stage… (P32L97)

The rewarding aspect of completing challenges sufficiently motivated and engaged players to continue playing, whilst also aiding the retention of information.

…when you finished a challenge you felt a great deal of enjoyment and a great deal of satisfaction in how your knowledge…caused you to proceed to another task. For example, there was a question on the crossword and you had to figure it out through clues that the fictional character had given you, like the newspaper, and you had to put it all together and find the shaded letters…I think that it's a good way to test your knowledge on the 1916 Easter Rising (P7L24)

…when you completed a challenge you want to go on like, "I can do this again, I know what to do now…" I wanted to strive to get to the end. The second crossword was a lot more challenging than the first one and we were coming to the end and time was running out and we were saying "there's no point in going on anymore…" – but when we started to get into it and getting the answers, we kept at it and kept on going (P9L325)

The diversity of the challenges was important in retaining player interest, bringing "freshness" to the game.

…Its not like every single challenge is the same…a lot of different variety…I thought it was a good way to keep the game fresh… (P9L328)

Furthermore, the inclusion of appropriate challenges were helpful in enriching player knowledge.

…they (the challenges) helped because they all had something to do with the course…and most of them were helpful in that they went to every part of the course that you needed to know about and they weren't just focused on one area – they were focused on different areas (P23L109)

4.1.3 Interim Assessment Evaluation Criterion

Presented below is a proposed evaluation criterion concerning Assessment.

A criterion for judging the educational quality of an ARG is the presence of appropriate and diverse in-game assessments that test the players' knowledge. Players must overcome assessments in order to progress in the game.

4.1.4 Player responses to proposed Assessment Evaluation Criterion

Each of the twenty-three players who completed Questionnaire E, agreed with the interim evaluation criterion on Assessment. Outlined below are a number of statements that the players made in the questionnaires and/or in the focus groups when the interim evaluation criterion was fed back to them.

Appropriate challenges and assessment helps players focus. This is beneficial to player learning and understanding.

…because the challenges were about the 1916 Rising it kept us focused on what we were learning…(EP7L73)

The assessment gives you a focus and an understanding of what the game is about…I feel its more beneficial when the game is focused on a fixed subject (EP32L71)

Assessment forced players to question and analyse what they were learning. This helped with the retention of information.

Without any assessment, we would have just flown through the game. We wouldn't have an excuse to stop and think about what we were doing. There'd be no challenge…no benefit to the game without assessment. Stopping and thinking actually meant that we were retaining information…if there's nothing in the game telling you that you're right or you're wrong then you're not learning at all…(EP8L76)

A variety of assessment and challenges reduces boredom –

I think it was good there was a variety of questions or tasks you had to do. You'd get bored with similar question after question… (EP9L39)

Overcoming assessments and challenges offered rewarding and motivational experiences –

…when you'd finally getting it (the assessment) right, the sense of achievement relieves the frustration…it kinda keeps the answer in your head for when you go back in a test, "oh this is the one I always got wrong, but now I know the right answer is…". Getting the assessments right was like a reward…(EP29L57)

…the assessment…challenged you to understand and research information on whatever the game was based on. Without this information you weren't able to progress in the game, so it definitely motivated you to find the answers... (EP32L73)

4.1.5 Assessment Evaluation Criterion

The incorporation of appropriate assessment, reflective of the subject matter of the ARG, is a characteristic of *Plunkett's Pages* that players felt was beneficial to their learning. Assessment allows players to gain an indication of their knowledge of the subject matter, tests their proficiency of the material, and motivates them to enhance their learning. A variety of assessment techniques can help sustain engagement.

The final evaluation criterion is presented as follows:

A criterion for judging the educational quality of an ARG is the presence of appropriate and diverse in-game assessments. Appropriate assessment ought to test the players' knowledge of the subject matter. Diverse assessment helps to reduce boredom. Assessment acts as a gateway for players to progress through the game, and ought to be reflective of the learning outcomes, motivate players to learn to succeed, aid information retention, and reward players for their knowledge.

Evaluation Criterion 1: Assessment

4.2 Sub Theme: Feedback

4.2.1 Introduction to Feedback

The second sub theme deals with the aspect of Feedback. In games, players are often presented with feedback that gives them insights into the nature of their performance.

4.2.2 Findings and Interpretation of Feedback

Feedback played a primary role in *Plunkett's Pages'* players' ability to gauge their performances. In cases where the players were incorrect with some of their assessment answers, the feedback was a motivational tool that benefited learning.

…sometimes the answers weren't right but it got you to keep looking for more (P9L213)

…it was good because…lets just say that you're doing a test and you do ten questions, and you might skip three or four, you'll never know the answers to them – but in the game you'll have to know the answer before you can move on and there's no other way around it… (P26L234)

…you'd be putting in the answers that you thought were right but you'd learn more finding out you were wrong, from your mistakes – the feedback helped you learn (P28L43)

4.2.3 Interim Feedback Evaluation Criterion

Presented below is a proposed evaluation criterion concerning Feedback.

A criterion for judging the educational quality of an ARG is the inclusion of a feedback system that notifies the players of their performance in a game.

4.2.4 Player responses to proposed Feedback Evaluation Criterion

The players outlined the benefits of Feedback as a learning aid –

…even to know when you got something wrong that it was wrong, allowed you to correct it with the right answer (EP29L57)

I think the feedback system is important because when I'm playing I like to know how I'm doing in the game and the way it let us know after each question if I got it wrong or I got it right helped me to take in the correct information (EP30L51)

Feedback was also an important motivational aid –

The immediacy of the feedback in the game means that you have to find the correct answer (EP22L37).

Finally, players argued that feedback needed to be positively constructive in their understanding of the subject material –

I think the feedback system was good as it alerted the player whether they were right or wrong. But if you did get it wrong the player wasn't insulted…you were helped to better understand the question (EP9L56).

4.2.5 Feedback Evaluation Criterion

The creation of a system whereby players are aware of their game performances in a Serious ARG is a desirable feature. At all times, feedback must be positively constructive and conveyed to the players, motivating them to continue playing.

A criterion for judging the educational quality of an ARG is the inclusion of a feedback system that immediately notifies and informs the players of their progress and performance in-game. Constructive feedback can motivate players to learn and understand the subject material more effectively.

Evaluation Criterion 2: Feedback

4.3 Sub Theme: Collaborative Learning

4.3.1 Introduction to Collaborative Learning

A feature of ARGs is collaborative play where players collectively pool their resources together to decipher clues, overcome challenges, piece together a fragmented narrative and work in unison to reach the game's conclusion. Collaboration, as distinct from competition, is the dominant form of game play in ARGs – but not necessarily in other game genres. This point was previously outlined in Table 1.

In Serious ARGs, collaborative play can lead to collaborative learning.

4.3.2 Findings and Interpretation of Collaborative Learning

The players were fully appreciative of the collaborative ethos of the ARG. Extoling the virtues of collective play, the players outlined the value of playing in a group and negating competitiveness.

…I did learn a lot. It shows you how much could be achieved when you work as a team because if you were to play this game on your own you wouldn't proceed very far (P9L87)

The players also felt that they benefited from the relaxed environment, where collectively they gained a better understanding of the material.

…you learn an awful lot more when you are playing with people you know and you weren't being put on the spot and being asked a question by the teacher. You had the help of your team and together you understood the questions better…got a better grasp of things (P21L39)

Learning was enhanced through open communication in a helpful, peer-centred environment. Collaborative play enhances the fun element of the game, and also augments focus and attentiveness amongst the playing group.

I felt it (teamwork) was good – you got to talk to your mates and find out how they did it and they could help you as well and you got to discuss how you did it (P8L32)

I think this is a lot better than classroom learning. You're having fun and you're not just sitting there saying "I just want to get this over and done with"… you pay attention more, and you don't be daydreaming, or talking to people…like in the classroom, you just have to listen and not talk, but in this you can talk to people about what is going on and it keeps you more focused (P18L447)

The active and sharing aspects of the Serious ARG, where the open lines of communication encouraged players to collaborate, was far removed from the typical "boredom" they associated with the traditional classroom.

…you could be just sitting there in a boring classroom and you wouldn't take in as much information as you need, but by playing the game with your friends you learn more, and you're more alert - you gather and share similar information, and you remember what you just done instead of sitting down with a textbook…. (P17L331)

I thought we worked well asking around a bit…even on our own Facebook where we shared different passwords and email addresses with each other… (P31L71)

Ultimately, players acknowledged that it was the sharing, collaborative nature of the participants play that eventually brought success to two groups.

There was a couple of times when we were really stuck on…then you'd the idea to get us into a big group and discuss it with everybody and I think everybody contributed to that…it helped us move forward. We might not have got so far playing the game on our own (P28L80)

Despite some initial bouts of competitiveness, the more successful groups quickly realised that the key to reaching their goals was to work collectively together. Over the course of the three days, the two groups who did complete the game did so by working collaboratively.

4.3.3 Interim Collaborative Learning Evaluation Criterion

Presented below is a proposed evaluation criterion concerning Collaborative Learning.

A criterion for judging the educational quality of an ARG is the presence of a collaborative play and shared learning ethos where players can openly communicate and learn from the game and their peers.

4.3.4 Player responses to proposed Collaborative Learning Evaluation Criterion

The players agreed with the interim evaluation criterion on Collaborative Learning, although some raised points about the aspect of competitive play.

Players espouse the collaborative benefits of playing and learning together with a shared goal.

…working collaboratively was beneficial to me because the other members of your team were there to support and help you to solve a problem. They might have picked up on something that you may have missed out on so it increased the knowledge of the team (EP32L115)

However, for some, the development of competition within the play environment can be motivational –

I liked the fact that you could play with each other cuz it helped if you were stuck. But I would like to see some competitiveness in the game as it encouraged players to finish quickly… (EP9L68)

I would prefer to work in a team against a team. I feel this would be more beneficial. As the game progresses, trying to solve the problems become more significant, and more is at stake – you wanna win and you don't wanna lose to another team…competition challenges you more and it motivates you to win, you don't wanna lose to another team so you work harder… (EP32L118)

Other players, however raised concerns with introducing competition into the learning environment –

…if competition was brought into the game, we would have really rushed through the game. Sometimes you have to stop and think – but with competition you wouldn't…you'd just find things here and there and skip other things… (EP20L115)

…if I was just being competitive with someone else I'd just rush through it, I'd wanna beat them and I wouldn't learn as much. Even if I got the right answer it would go out of my head straight away cuz I'd be moving onto the next question… (EP23L75)

…with competition you'd be rushing against someone and it could backfire…you couldn't gain or learn from each other (EP26L44)

4.3.5 Collaborative Learning Evaluation Criterion

The collaborative learning nature of *Plunkett's Pages* provided a new experience for the players. Working together towards a shared goal, the players were able to enjoy the relaxed learning environment that lifted the burden from individuals. Collectively, players learned from the game and each other.

While collaboration (as distinct from competition) is usually considered the dominant form of game play in ARGs, some small-scale competitive elements can invoke player motivation. Reaching a balance in this collaborative/competitive scale is an important consideration.

A criterion for judging the educational quality of an ARG is the presence of dominantly collaborative play and a shared learning ethos. Competition is discouraged as the dominant mechanism. However, a small element of competitive play is acceptable. In a collaborative environment, where communication is open, players can support and be supported, to learn from the game and their peers.

Evaluation Criterion 3: Collaborative Learning

4.4 Sub Theme: Debriefing

4.4.1 Introduction to Debriefing

A debriefing session in the aftermath of an entertainment ARG does not normally take place. This is primarily due to TINAG. However, a post-game reflection period at the conclusion of a Serious Game is a typical occurrence.

4.4.2 Findings and Interpretation of Debriefing

Debriefing is an important aspect in bookending the narrative and game play lessons that players derive from playing a Serious ARG. The blurring of the lines of reality and fiction in an ARG can lead to confusion. In *Plunkett's Pages,* players recognised the importance of the debriefing session in pointing out the factual and fictional elements of the game.

I thought Lucy Reilly was real. I couldn't really distinguish between what was fact and what was fiction and who was real. The debriefing session described it really well (P13L153)

You could believe that Lucy Reilly was real and that some characters weren't fictional (P19L151)

Being able to distinguish between the factual and fictional elements of the ARG was an important factor for the players, especially as they would be tested on what they were learning in a state exam, just months later.

It is important to know the difference because the Easter Rising will come up in our Junior Cert and we'll know now that Tom Reilly and Lucy Reilly are fictional characters and they weren't in the Easter Rising… (P30L226)

Debriefing is important too in reducing any of the complexities associated with the game, and outlining any aspects of the game play that the players may not have experienced or may have misunderstood.

I think that when you were showing us going back over it as well it helped us see how complicated and complex the game really was…its not just straightforward…even when we were getting the answers right with the videos and the audio, there was far much more to do… (P30L216)

4.4.3 Interim Debriefing Evaluation Criterion

Presented below is a proposed evaluation criterion concerning Debriefing.

A criterion for judging the educational quality of an ARG is the presence of a debriefing session in the immediate aftermath of the game that provides adequate time for post-game analysis, and helps distinguish between factual and fictional elements.

4.4.4 Player responses to proposed Debriefing Evaluation Criterion

The players agreed with the interim evaluation criterion on Debriefing outlining its benefits as a means of delineating fact from fiction.

The Serious ARG successfully blurred the lines of reality and fiction. Therefore, the debriefing session was an important exercise for all players –

…the characters were created so real that you would have believed that they were a part of it all. It was important to know that they all weren't (EP7L129)

The characters were made out to be so believable that without the debriefing all my knowledge would have been mixed up (EP29L134)

I think the debriefing sessions are very important after the game because the realism of the game may lead you to believe that some of the characters exist especially when the video and audio are there to confirm that the characters are actually human. I think it is important after the game to know the difference between the fact and the fiction for future references and for the subject of the game…at the time we played the game we were going into our Junior Certificate so I felt it was important that we knew where the facts stop and the fiction started (EP32L143)

4.4.5 Debriefing Evaluation Criterion

Based on the researcher's experience of running *Plunkett's Pages,* game creators and educators are advised to conduct a debriefing session at the conclusion of any Serious ARG. This session should discuss the merits of the game play, its narrative, challenges and learning, and detangle the factual and fictional elements of the ARG. This process can help ascertain if and where the learning occurred, and if and where improvements can be made in future games.

A criterion for judging the educational quality of an ARG is the presence of a debriefing session in the immediate aftermath of the game that provides adequate time for post-game analysis. Debriefing reflects on the play and learning from a Serious ARG, helping improve its overall quality, and demarcates the factual and fictional elements of the game.

Evaluation Criterion 4: Debriefing

5.0 DISCUSSION

In Serious ARGs, education and knowledge transfer is of paramount importance. The quantitative results from Questionnaire A indicate that *Plunkett's Pages* succeeded in educating the players. The alignment of learning outcomes and game play, particularly through assessments and challenges, is central to achieving the objective of educating players (de Freitas 2006, McCall 2011, Whitton 2011a). Open dialogue between the researcher and the educators during the planning phase helped align the learning outcomes with the assessments and challenges in *Plunkett's Pages.* Linking a game and its challenges with its learning outcomes, assessment or formal education is a complex matter. ARGs are no different. Challenges are a timely reminder to players of their game performance and form the basis for problem-based learning to occur in the ARG. The implementation of appropriate assessment within the context of the ARG is beneficial to the learning process for players *(Evaluation Criterion 1)*.

Performance indicators, stemming from assessment, are beneficial to the players' learning. Feedback effectively enhances the learning process for players, providing them with an immediate representation of their game play performances *(Evaluation Criterion 2)*. In *Plunkett's Pages,* feedback was always constructive, reassuring players at times when they felt challenged or underwhelmed (EP7L91, EP9L36). The development of such a feedback system also acts as a motivational tool (Learning and Teaching in HE 2009, Markovic et al. 2007, Moseley 2009). Regular responses encourage and engage players to be passionate about learning in a more effective and efficient manner (Prensky 2001). The absence of a feedback system detaches players from the learning and discourages them from devoting time to play and learn.

The support of the collective in ARGs allows players to carry out more complex tasks than they would as an individual. This collective ethos aligns itself with the "hive mind" of ARGs (ARGology 2010, McGonigal 2007) and the participatory culture of learning in groups (Bransford et al. 1999, Roschelle et al. 2001). Collaborative play allows players to gain insightful knowledge in a game-like engaging environment that presents opportunities for effective independent and collective learning. By their own admission, the collaborative play in *Plunkett's Pages* was the primary reason why some succeeded (P28L80) in finding the whereabouts of the diary. The collaborative sharing nature of *Plunkett's Pages* draws parallels with the peer support feature many ARG communities, including *Perplex City,* experienced. This is key to enhancing the learning within ARGs (Moseley 2008). Instigating a collective game play ethos inspires players to collaboratively learn together in their communities *(Evaluation Criterion 3),* and is central to the success of any ARG or Serious ARG. Collaboration, as distinct from competition is usually considered the dominant form of game play in ARGs. However, as seen in *Plunkett's Pages,* some small-scale competitive elements can invoke player motivation. Reaching a balance in this collaborative/competitive scale is an important consideration.

In Serious Games, learning needs to be re-affirmed in post-game analysis and reflection exercises to ensure that the "correct embedding of the virtual experience" takes place (de Freitas 2006, p. 11). A lack of post-Serious ARG discussion can leave players unaware of aspects of the game they may have missed out on or may not entirely detangle the factual and fictional elements of the game. Debriefing plays an integral role in the learning in educational games (Crookall 2010, de Freitas 2006, Kriz 2010, Leigh 2003). Learning comes from, and happens within the debriefing (Crookall 2010), and as such, integrating the post-game reflection exercise into the learning context is vitally important. However, the provision for reflection or debriefing is not something that is typically associated with ARGs. For one, allowing post-game discussions defies the ethos of TINAG. Yet, by not allowing players to discuss the game in its aftermath, player learning is severely impacted. Debriefing took place in the immediate aftermath of *Plunkett's Pages* and allowed the players to collectively reflect on their experiences *(Evaluation Criterion 4)*. The intention of the debriefing was to provide clarity on any aspects of their game play, to outline areas where they might have struggled, and to answer any questions the groups may have had. One further aspect of the debriefing was to explicitly outline and document the factual and fictional elements of the Serious ARG. This was important in ensuring that the players were aware of accurate depictions of the narrative and game play (McCall 2011). Without debriefing, the full potential for learning is curtailed.

6.0 CONCLUSION

Plunkett's Pages provided an interactive narrative game that supplemented an active, GBL experience for its players, and allowed them to become knowledgeable about the events of Easter 1916. This study has provided many lessons, with those presented in this article predominately pertaining to how to use an ARG to invoke learning. Firstly, potential game designers and educators share a mutual responsibility to agree on the appropriateness and diversity of in-game challenges, and plan these in accordance with the games' learning outcomes. Secondly, feedback is a useful mechanism in delineating player performances, and an effective motivational technique in enhancing the learning process. Thirdly, collaboration can bring great satisfaction and a sense of collective achievement to players, allowing them to discuss game instances and share information with their peers in ways they might not normally encounter. Finally, the study has highlighted the merits of debriefing, and the absolute necessity for post-game reflective discussions to take place in order to disentangle factual and fictional elements and complement the learning experience.

A step away from the passivity that learners may associate with the traditional classroom, the dynamic nature of Serious ARGs can motivate players by giving them a focused problem to solve. The solution process elicits knowledge and learning rewards, while goal-driven environments morph player game play into a focused, engaged learning experience. Lessons from such case studies should help ARGs reach their potential as a GBL tool.

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